

Unghui Nam

Graduate Student, Seoul National University

Research Interests: Robot Learning, Physical AI

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About Me

I am a graduate student working on **robot learning** and **physical AI**, advised by Prof. Byonghyo Shim at Seoul National University. My research focuses on building reliable and efficient learning systems for robots, with the goal of designing autonomous robotic systems that can perceive, reason, and act through interaction with the physical world.

Education

Seoul National University

Mar. 2020 – Feb. 2026

B.S. in Mechanical and Aerospace Engineering, School of Engineering

Seoul National University

Mar. 2026 – Present

M.S.&Ph.D. in Electrical and Computer Engineering

Publications

** denotes equal contribution.*

[1] **Adaptive Capacity Allocation for Vision Language Action Fine-tuning**

D.H. Kim, M.J. Bae*, U. Nam*, G.H. Kim*, S.Y. Lee*, K.H. Shim, B. Shim

IEEE International Conference on Robotics and Automation (ICRA), 2026

Industry Experience

Data Analyst

Jan. 2023 – Jun. 2023

Whatssub Inc.

- Designed payment-transaction categorization schemes and built an automated classification pipeline.
- Developed an LLM-based system that generates personalized marketing copy for target user segments, and integrated it into an automated production workflow.

Selected Projects

Multinational ALPR in Cameras with Low Resolution and Low Frame Rate

Sep. 2024 – Dec. 2024

Course Project, Fundamentals of Deep Learning, SNU

- Surveyed and reimplemented existing license-plate detection and recognition models.
- Led benchmark evaluation on the dataset and authored the final technical report.

Automated Generation of Paint-by-Number Templates

Sep. 2024 – Dec. 2024

Course Project, Computer Vision, SNU

- Designed and implemented edge-detection, blurring, and color-clustering modules for an automated paint-by-number template generator.

- Conducted performance evaluation of the end-to-end pipeline.

Zumo Shield Robot

Sep. 2024 – Dec. 2024

Course Project, Aerospace Sensor Systems, SNU

- Implemented line-tracing and object-avoidance behaviors using infrared and ultrasonic sensors.
- Developed a self-localization module fusing gyroscope and accelerometer measurements.

Piuda Project – SW Development Competition

May 2024 – Jul. 2024

Project Lead & Designer, App for Supporting Homeless Individuals

- Led overall application planning and UI/UX design.
- Advanced to the finals of the competition.

Technical Skills

Programming Python, C++

Robotics ROS 2

Interests Robot Learning, Vision-Language-Action Models, Physical AI

Languages Korean (Native), English (Fluent)

Certifications & Language Tests

Craftsman Information Processing National Technical Qualification

Apr. 2023

TOEFL iBT Score: 104

Sep. 2016